Weil Conjectures

Chapter 15: Pure sheaves on curves

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Contents

15 Pure sheaves on curves 1

15.1 Weights of middle extensions 2

15.2 Growth of squared traces 6

Outside this export 10

References 11

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15 Pure sheaves on curves

How much does the trace function of a sheaf know about its Frobenius eigenvalues? For a sheaf of rank one, pure of weight w, on a dense open U_0 of the line, it knows the weight. Indeed, every trace over $\mathbf{F}_{q^n}$ has absolute value $q^{nw/2}$, so the sum S_n of the $|t(x)|^2$ over $U_0(\mathbf{F}_{q^n})$ is $q^{nw}\#U_0(\mathbf{F}_{q^n})$, about $q^{n(w+1)}$, and the growth rate q^{w+1} of these sums therefore determines w. In higher rank, however, a trace is a sum of eigenvalues, and the eigenvalues can cancel. For example, on $\mathcal{B}_b \oplus \mathcal{B}_{-b}$ the trace at a point of degree n is $b^n + (-b)^n$, which is zero for every odd n. We show that the cancellation cannot persist: for a nonzero pure lisse sheaf of weight w on an affine curve, $\limsup_n \frac{1}{n}\log_q S_n = w+1$ (Theorem 15.14).

The reason is positivity once more. Recall that a sheaf $\mathcal{G}$ is real if the characteristic polynomials of Frobenius on its stalks have real coefficients under ι (Definition 14.21*). For such $\mathcal{G}$, pure of weight w, $|t_{\mathcal{G}}|^2$ is the trace function of $\mathcal{G}\otimes\mathcal{G}$, and hence the generating series of the S_n is the logarithmic derivative of $L(\mathcal{G}\otimes\mathcal{G},t)$. Positivity of the Euler factors (Lemma 14.22*) shows that $L(\mathcal{G}\otimes\mathcal{G},t)$ is not a polynomial, and Lemma 14.20* puts its poles on $|t| = q^{-w-1}$. Together these force the sums S_n to reach the size $q^{n(w+1)}$ infinitely often. The restriction to real sheaves costs nothing, since an arbitrary pure sheaf becomes real after a twist to weight zero and the addition of its dual (Proposition 15.9).

This is what the Fourier argument of Chapter 17* needs. That argument starts from a pure lisse sheaf M of weight w on a dense open $j : U_0 \hookrightarrow \mathbf{A}^1$ and produces a pure lisse sheaf M' on a dense open j' of the dual line. It knows the trace function of M' exactly, as the discrete Fourier transform of that of M, but it cannot see the weight of M' directly. Since Plancherel's formula multiplies the sums S_n by q^n, the theorem shows that M' has weight $w+1$.

The group we finally want to bound, $H^1_c(\mathbf{A}^1, j_*M)$, is the stalk of j'_*M' at the origin. But the origin may be one of the points where M' is not lisse, and there we need to know that j'_* does not create larger weights (Proposition 15.5).

15.1 Weights of middle extensions

Let j be the inclusion of a dense open in a smooth curve, and let $\mathcal{F}$ be a lisse sheaf on the open. The Fourier argument needs an extension of $\mathcal{F}$ to the whole curve that loses nothing at the boundary points. The ordinary direct image $j_*\mathcal{F}$ is such an extension, whereas extension by zero loses something. For example, for the constant sheaf on $\mathbf{G}_m \subset \mathbf{A}^1$, $j_!\overline{\mathbf{Q}}_\ell$ has stalk 0 at the origin, while the constant sheaf on $\mathbf{A}^1$ is the natural extension. In general, by Lemma 15.2, $j_*\mathcal{F}$ recovers a lisse extension whenever one exists, and its stalk at a boundary point is the space of inertia invariants.

If $\mathcal{F}$ is punctually of weights $\leq a$, then so is $j_*\mathcal{F}$, at the boundary points as well (Proposition 15.5). In Chapter 17* this bounds the weights of the boundary terms in the Fourier transform of a middle extension.

Definition 15.1 (Middle extension)**.** Let $j : U \hookrightarrow X$ be a dense open in a smooth curve and let $\mathcal{F}$ be a lisse sheaf on U. The *middle extension* of $\mathcal{F}$ is the ordinary direct image $j_*\mathcal{F}$ of sheaves.

Lemma 15.2 (The boundary stalk)**.** *Let $j : U_0 \hookrightarrow X_0$ be a dense open in a smooth curve, let $\mathcal{F}$ be a lisse Weil sheaf on U_0, and let j_* be the ordinary direct image of sheaves. Let $\bar{z}$ be a geometric point over a point z of $X_0 - U_0$, and let I_z be the inertia group at z, acting on the local generic fibre V of $\mathcal{F}$. There is a canonical Frobenius-equivariant identification*

$$(j_*\mathcal{F})_{\bar{z}} = V^{I_z}.$$

If $\mathcal{F}$ extends to a lisse sheaf on X_0, the adjunction map from that extension to $j_*\mathcal{F}$ is an isomorphism.

Proof. We prove the formula for finite coefficients, where it is Galois theory of the punctured henselian neighbourhood, and pass to the limit.

The stalk of $j_*\mathcal{F}$ at $\bar{z}$ is the colimit of the sections of $\mathcal{F}$ over the inverse images of U in étale neighbourhoods of $\bar{z}$, that is, the group of sections over the punctured spectrum of the strict henselization at $\bar{z}$. This punctured spectrum is the spectrum of a field, whose absolute Galois group is the inertia group I_z, so finite locally constant sheaves there correspond to finite continuous representations of the inertia group, and their sections are the invariant vectors. Applying this to the finite reductions of a stable lattice and passing to the inverse limit, we obtain the formula with integral coefficients.

Two compatibilities take us to rational coefficients and to $\overline{\mathbf{Q}}_\ell$. Rationalization commutes with taking invariants, since a power of the uniformizer carries an invariant rational vector into the lattice. A finite extension of scalars also commutes with invariants. Indeed, it is exact, hence commutes with finite intersections of kernels, and in a finite-dimensional space the intersection of the kernels of all $g-1$ is the intersection of finitely many of them. This gives the formula with rational coefficients, after finite extension of scalars, and hence also over $\overline{\mathbf{Q}}_\ell$. Since all these maps commute with the action of the decomposition group, the identification is Frobenius equivariant. Finally, if $\mathcal{F}$ extends to a lisse sheaf on X_0, the local inertia acts trivially, and so the adjunction map is an isomorphism on every geometric stalk. ∎

Example 15.3 (Boundary stalks on $\mathbf{G}_m \subset \mathbf{A}^1$)**.** Let $j : \mathbf{G}_{m,k_0} \hookrightarrow \mathbf{A}^1_{k_0}$ and $z = 0$. The covers $u^n = t$ are totally ramified at 0, so the inertia I_0 maps onto the monodromy of a Kummer sheaf. For the Kummer sheaf of a nontrivial character, I_0 acts on the line V through that character. Hence $V^{I_0} = 0$, and the stalk of j_* of this sheaf at $\bar{0}$ is zero, so here j_* and $j_!$ agree. For the

Kummer extension $\mathcal{E}$ of Example 13.4*, on the other hand, I_0 acts by the matrices $\left(\begin{smallmatrix}1 & \kappa \\ 0 & 1\end{smallmatrix}\right)$ with κ running over $\mathbf{Z}_\ell(1)$, whose common fixed vectors form the line $\overline{\mathbf{Q}}_\ell(1)$. So the stalk of $j_*\mathcal{E}$ at 0 is $\overline{\mathbf{Q}}_\ell(1)$, of weight -2, while $\mathcal{E}$ has weights -2 and 0; the stalk at the boundary keeps only part of the weights.

To bound $j_*\mathcal{F}$ we compare the L-functions of $\mathcal{F}$ and of $j_*\mathcal{F}$ on a disc. To see that the Euler product converges there, we need a crude bound for the number of points of a curve.

Lemma 15.4 (A point-count bound)**.** *For a smooth affine curve U_0/k_0 there is a constant D such that $\#U_0(k_n) \le Dq^n$ for all $n \ge 1$. For a finite k_0-scheme, the number of k_n-points is bounded independently of n.*

Proof. We map U_0 finitely and flatly onto the affine line, whose number of k_n-points is q^n, and bound the size of the fibres.

By Noether normalization [Stacks, 00OY] there is a finite map $U_0 \to \mathbf{A}^1_{k_0}$, and the coordinate ring of U_0 is a finite torsion-free module over $k_0[t]$. To see this, let $f \neq 0$ in $k_0[t]$ and $fa = 0$. On each irreducible component of U_0 the image of f is nonzero, since otherwise the component would be finite over $k_0[t]/(f)$ and hence of dimension zero. Since the coordinate ring of each component is a domain, $fa = 0$ forces a to vanish on every component, and then $a = 0$ because U_0 is reduced. As $k_0[t]$ is a principal ideal domain, it follows that the coordinate ring of U_0 is free of some rank D. Hence, after any extension of the base field, every fibre has a coordinate ring of dimension D, and by the Chinese remainder theorem it has at most D rational points. Summing over $\mathbf{A}^1(k_n)$, we obtain the first bound. The second holds because the number of rational points of a finite scheme is at most its number of geometric points. ∎

In the next proof we use the lemma to show that the logarithm of the Euler product of $\mathcal{F}$ converges on a disc of radius $q^{-a/2-1}$, and on this disc we compare it with $j_*\mathcal{F}$.

Proposition 15.5 (Weights of j_* at the boundary points)**.** *Let $j : U_0 \hookrightarrow X_0$ be a dense open of a smooth geometrically connected curve. If a lisse Weil sheaf $\mathcal{F}$ on U_0 is punctually of weights $\le a$, so is $j_*\mathcal{F}$ on X_0.*

Proof. Comparing the L-functions of $\mathcal{F}$ and of $j_*\mathcal{F}$ gives the weaker bound $a+2$ at the boundary, and tensor powers remove the extra 2.

The assertion is local at a point of $X_0 - U_0$, so we may replace X_0 by an affine neighbourhood of that point. The complement of U_0 is an effective Cartier divisor, so after shrinking this neighbourhood it is cut out by one equation f, and then $U_0 = D(f)$ is affine as well. Write $i : Z_0 = X_0 - U_0 \hookrightarrow X_0$. The hypotheses are preserved by this replacement. Indeed, forming j_* commutes with restriction to open subschemes, and the restriction of $\mathcal{F}$ is still punctually of weights $\le a$. Moreover, a nonempty open of X_0 is still geometrically connected.

The boundary Euler factors of $j_*\mathcal{F}$ are the ratio L_X/L_U of the two L-functions defined below. We show that L_U has no zero and L_X no pole on the disc $|t| < q^{-a/2-1}$, so that the boundary factors have no pole there, and this bounds their eigenvalues.

The L-function on U_0. Put $L_U(t) = L(U_0, \mathcal{F}, t)$. We show that ιL_U is holomorphic and nonvanishing on the disc, by bounding the coefficients of its logarithm. The logarithm of the Euler product has coefficients

$$\log \iota L_U(t) = \sum_{n \ge 1} \frac{c_n}{n} t^n, \qquad c_n = \sum_{d \mid n} d \sum_{\deg(x) = d} \iota \operatorname{Tr}(\operatorname{Frob}_x^{n/d} \mid \mathcal{F}_{\bar{x}}).$$

Let $r = \operatorname{rank} \mathcal{F}$. For a closed point x of degree $d \mid n$, the bound on the weights gives $|\iota \operatorname{Tr}(\operatorname{Frob}_x^{n/d} \mid \mathcal{F}_{\bar{x}})| \leq r q_x^{(n/d)a/2} = r q^{na/2}$. Since $\sum_{d|n} d \cdot \#\{x \in |U_0| : \deg(x) = d\} = \#U_0(k_n)$, Lemma 15.4 gives $|c_n| \leq r D q^{n(a/2+1)}$. Hence the logarithm converges on the disc $|t| < R = q^{-a/2-1}$, and its exponential is a nonvanishing holomorphic function there. By the trace formula, its Taylor series is that of the rational function $\iota L_U(t)$, so the two agree on the whole disc.

The L-function on X_0. Let $L_X(t) = L(X_0, j_*\mathcal{F}, t)$. We show that ιL_X is holomorphic on the same disc. By the trace formula its poles come from H_c^0 and H_c^2 of $j_*\mathcal{F}$, and we compute both groups.

The group H_c^0 vanishes. Indeed, the support of a compactly supported section of $j_*\mathcal{F}$ on X is closed and proper over k, and since X is an affine curve, the support is finite. So the section vanishes on a dense open subset of U, and as $\mathcal{F}$ is lisse and U is connected, it vanishes on all of U. Since a section of $j_*\mathcal{F}$ is determined by its restriction to U, the section is zero. For H_c^2, note that the quotient sheaf in

$$0 \longrightarrow j_!\mathcal{F} \longrightarrow j_*\mathcal{F} \longrightarrow i_*i^*j_*\mathcal{F} \longrightarrow 0$$

is supported on the finite boundary, whose cohomology is concentrated in degree zero. From the long exact sequence we obtain $H_c^2(X, j_*\mathcal{F}) = H_c^2(U, \mathcal{F})$, and by Lemma 14.20* its eigenvalues have absolute value at most $q^{a/2+1}$. By the trace formula, every pole of ιL_X is the inverse of a Frobenius eigenvalue on H_c^0 or H_c^2. Since $H_c^0 = 0$, the poles have absolute value at least $q^{-a/2-1} = R$. Hence ιL_X is holomorphic on $|t| < R$.

The boundary factors. On U_0 the sheaves $\mathcal{F}$ and $j_*\mathcal{F}$ have the same stalks, so the two Euler products differ only by the factors at the points of Z_0, and we get

$$L_X(t) = L_U(t) \prod_{z \in Z_0} \det(1 - \operatorname{Frob}_z t^{\deg(z)} \mid (j_*\mathcal{F})_{\bar{z}})^{-1}.$$

Since L_U has no zero on the disc, the finite product on the right is holomorphic there. As it is the reciprocal of a product of polynomials with constant term one, none of these polynomials has a zero on the disc. Now, for an eigenvalue α at z, the roots of $1 - \iota(\alpha)t^{\deg(z)}$ have absolute value $|\iota(\alpha)|^{-1/\deg(z)}$. This number is therefore at least R, which is the preliminary bound

$$|\iota(\alpha)|^2 \leq q_z^{a+2}.$$

Tensor powers. The error 2 in the exponent comes from the twist (-1) on H_c^2. Since it does not grow with the weight, it becomes negligible for high tensor powers. We therefore apply the preliminary bound to $\mathcal{F}^{\otimes m}$ for every $m \geq 1$, a sheaf punctually of weights $\leq ma$. Let V be the local generic fibre of $\mathcal{F}$ at z. By Lemma 15.2, the eigenvalue α of Frob_z on $(j_*\mathcal{F})_{\bar{z}}$ has an eigenvector $v \in V^{I_z}$, and then $v^{\otimes m}$ is a nonzero invariant eigenvector in $V^{\otimes m}$ with eigenvalue α^m. By the same lemma, α^m is therefore an eigenvalue on the stalk of $j_*(\mathcal{F}^{\otimes m})$ at $\bar{z}$, and the preliminary bound for $\mathcal{F}^{\otimes m}$ gives $|\iota(\alpha)|^{2m} \leq q_z^{ma+2}$. Taking mth roots and letting m tend to infinity, we obtain $|\iota(\alpha)|^2 \leq q_z^a$ at every boundary point. On U_0, finally, the bound is the hypothesis. ∎

The proposition gives only an upper bound at the boundary, and in general nothing better is true. For example, for a family of elliptic curves with multiplicative reduction at z, the first cohomology of the fibres is pure of weight 1, while its inertia invariants at z form a line of weight 0. When the inertia acts trivially, however, we get the lower bound by duality.

Corollary 15.6 (An unramified boundary point)**.** *Let $\mathcal{F}$ be a lisse sheaf, pure of weight a, on a smooth geometrically connected curve with a point removed. If the inertia at the removed point acts trivially, then the stalk at that point of the direct image of $\mathcal{F}$ to the curve is pure of weight a. In particular, a lisse Weil sheaf on a smooth geometrically connected curve that is pure on a dense open is pure on the whole curve.*

Proof. Purity follows from the upper bounds for the sheaf and for its dual, provided the stalk of the dual is the dual of the stalk, and trivial inertia guarantees this. So we apply Proposition 15.5 to the sheaf and to its dual, which is pure of weight $-a$. Since the inertia acts trivially, the stalk of the extended dual is the dual of the stalk of the extension, and so every eigenvalue α satisfies $|\iota\alpha|^2 \leq q_z^a$ and $|\iota\alpha^{-1}|^2 \leq q_z^{-a}$, and equality holds.

For a sheaf that is already lisse on the curve, the inertia is trivial, and Lemma 15.2 identifies the stalk of j_* with the original stalk. The first assertion, applied at each of the finitely many points outside the dense open, then gives the second. ∎

The proof of Proposition 15.5, together with Lemma 14.20*, also gives the extreme degrees of the cohomology of a middle extension on an affine curve.

Lemma 15.7 (The extreme degrees on an affine curve)**.** *Let $j : U_0 \hookrightarrow C_0$ be a dense open in a smooth geometrically connected affine curve, let F be lisse on U_0, and put $V = F_{\bar{\eta}}$. Then $H_c^i(C, j_*F) = 0$ for $i \neq 1, 2$, and there is a canonical identification*

$$H_c^2(C, j_*F) = V_{\pi_1(U)}(-1).$$

If F is pure of weight w, the group in degree two, when nonzero, is pure of weight $w + 2$.

Proof. The vanishing of H_c^0 was shown in the proof of Proposition 15.5. For the higher degrees, let i be the inclusion of $C_0 - U_0$. Since sheaves with finite support have compact-support cohomology only in degree zero, the localization sequence for $0 \to j_!F \to j_*F \to i_*i^*j_*F \to 0$ identifies $H_c^r(C, j_*F)$ with $H_c^r(U, F)$ for $r \geq 2$. Lemma 14.20* gives the value and the weight of this group for $r = 2$, and its vanishing for $r > 2$. ∎

For the perverse t-structure in Chapter 16* we need the same range of degrees for arbitrary constructible sheaves.

Lemma 15.8 (Cohomological range on a curve)**.** *Let C be a smooth curve over an algebraically closed field and let F be a constructible sheaf on C with coefficients in E or $\overline{\mathbf{Q}}_\ell$. Then $H_c^i(C, F) = 0$ for $i \notin \{0, 1, 2\}$.*

Proof. Choose a dense open $j : U \hookrightarrow C$ on which F is lisse, and let i be the inclusion of the finite complement. The sequence $0 \to j_!j^*F \to F \to i_*i^*F \to 0$ gives a long exact sequence in which $H_c^r(C, i_*i^*F)$ vanishes for $r \neq 0$, because the support is finite. It therefore suffices to show $H_c^r(U, j^*F) = 0$ for $r \notin \{0, 1, 2\}$. For $r < 0$ there is nothing to prove, and for $r > 2$, degreewise Poincaré duality (Corollary 10.13*, with $d = 1$) shows that $H_c^r(U, j^*F)$ is dual to $H^{2-r}(U, (j^*F)^\vee(1))$, which is zero because $(j^*F)^\vee$ is a sheaf and $2 - r < 0$. ∎

A constructible Weil sheaf on a curve C_0 over k_0 gives a constructible sheaf on $C = C_0 \times_{k_0} k$, so the lemma applies to it as well.

15.2 Growth of squared traces

In Chapter 17* we know the trace function of the Fourier transform M' exactly, but not its weight. For a pure sheaf the trace function determines the weight: it is the growth rate of the sums of squared traces (Theorem 15.14).

Proposition 15.9 (Twisting to weight zero and adding the dual)**.** *Let $\mathcal{F} \neq 0$ be lisse and pure of weight w on a nonempty finite-type scheme over k_0. There is $b \in \overline{\mathbf{Q}}_\ell^\times$ with $|\iota(b)| = q^{-w/2}$. Put*

$$\mathcal{F}^0 = \mathcal{F} \otimes \mathcal{B}_b, \qquad \mathcal{E} = \mathcal{F}^0 \oplus (\mathcal{F}^0)^\vee.$$

Then $\mathcal{E}$ is real and pure of weight zero, contains $\mathcal{F}^0$ as a direct summand, and at points over k_n its trace is twice the real part of the trace of $\mathcal{F}^0$.

Proof. A twist to weight zero puts all local eigenvalues on the unit circle, where inversion is complex conjugation. Adding the dual therefore makes each multiset of eigenvalues stable under conjugation.

To find the twist, choose a closed point x, of degree d, at which the stalk is nonzero, and a Frobenius eigenvalue α at x. Let $b \in \overline{\mathbf{Q}}_\ell$ be a dth root of α^{-1}. By purity, $|\iota(b)| = |\iota(\alpha)|^{-1/d} = q^{-w/2}$, and Proposition 13.7* shows that $\mathcal{F}^0$ is pure of weight zero.

Its local eigenvalues β_j therefore have absolute value one after applying ι. The local eigenvalues of its dual are the β_j^{-1}, and $\iota(\beta_j^{-1}) = \overline{\iota(\beta_j)}$. So the union of the two multisets is stable under complex conjugation, and $\mathcal{E}$ is real and pure. In the same way, at a point x over k_n the eigenvalues of $\mathrm{Frob}_{x,n}$ on $(\mathcal{F}^0)^\vee$ become, after ι, the complex conjugates of those on $\mathcal{F}^0$. Hence ι of the trace of $(\mathcal{F}^0)^\vee$ is the complex conjugate of ι of the trace of $\mathcal{F}^0$, and the trace of $\mathcal{E}$ is twice its real part. ∎

The number ν below is the growth exponent of the sums of squared traces, shifted by 1 so that it equals the weight. We take a lim sup because the sums can vanish for infinitely many n, as the example of $\mathcal{B}_b \oplus \mathcal{B}_{-b}$ at the start of this chapter shows.

Definition 15.10 (Growth of squared traces)**.** Let K be a constructible Weil complex on an affine curve X_0. For $n \geq 1$ and $x \in X_0(k_n)$, let $\mathrm{Frob}_{x,n}$ be the Frobenius acting on the stalk $K_{\bar{x}}$ (Theorem 12.24*), and set

$$t_{K,n}(x) = \sum_i (-1)^i \operatorname{Tr}(\mathrm{Frob}_{x,n} \mid \mathcal{H}^i(K)_{\bar{x}}), \qquad S_n(K) = \sum_{x \in X_0(k_n)} |\iota t_{K,n}(x)|^2.$$

Put

$$\nu(K) = \limsup_{n \to \infty} \frac{\log_q S_n(K)}{n} - 1, \qquad \log_q 0 = -\infty.$$

A cohomological shift changes the weight of a complex in the sense of Definition 13.1*. But it only multiplies the trace function by a sign, and so it leaves ν unchanged.

In the theorem below we find ν from the radius of convergence of a generating series, which is a logarithmic derivative of an L-function.

Lemma 15.11 (Radius of convergence)**.** *Let K be a constructible Weil complex on an affine curve X_0. The series $\varphi_K(t) = \sum_{n \geq 1} S_n(K) t^{n-1}$ has radius of convergence $q^{-\nu(K)-1}$, where this number is read as $+\infty$ if $\nu(K) = -\infty$ and as 0 if $\nu(K) = +\infty$.*

Proof. By the Cauchy–Hadamard formula the radius is the inverse of $\limsup_n S_n(K)^{1/n} = q^{\nu(K)+1}$; the shift of the index by one does not change it. ∎

To compute ν we also need upper bounds for lisse sheaves and for complexes with finite support, as well as the effect of twisting by $\mathcal{B}_b$.

Lemma 15.12 (Bounds for squared traces)**.** *Let X_0 be a smooth affine curve over k_0, and let a be a real number.*

1. *If F is a lisse Weil sheaf of rank r on X_0, punctually of weights $\leq a$, then $S_n(F) \leq Dr^2 q^{n(a+1)}$ for all n, with D as in Lemma 15.4, and hence $\nu(F) \leq a$.*
2. *If K is a constructible Weil complex on X_0 supported at finitely many closed points, and every eigenvalue α of Frob_x on every $\mathcal{H}^i(K)_{\bar{x}}$ satisfies $|\iota(\alpha)| \leq q_x^{a/2}$, then there is a constant C with $S_n(K) \leq Cq^{na}$ for all n, and hence $\nu(K) \leq a-1$.*
3. *For every constructible Weil complex K on X_0 and every $b \in \overline{\mathbf{Q}}_\ell^\times$, we have $\nu(K \otimes \mathcal{B}_b) = \nu(K) + \lambda_\iota(b)$, with $\mathcal{B}_b$ as in Proposition 13.7*.*

Proof. For (1), the trace at each point of $X_0(k_n)$ has absolute value at most $rq^{na/2}$, and Lemma 15.4 gives $S_n \leq Dr^2 q^{n(a+1)}$. For (2), Lemma 15.4 bounds the number of points of the support with values in k_n independently of n, so it suffices to bound the trace at each of them. Since the support consists of finitely many points with finite-dimensional stalks, the sum c of the ranks of the stalks of the $\mathcal{H}^i(K)$ is finite, and at each such point the trace has absolute value at most $cq^{na/2}$. For (3), the trace of $\mathcal{B}_b$ at a k_n-point is b^n, so tensoring by $\mathcal{B}_b$ multiplies S_n by $|\iota(b)|^{2n} = q^{n\lambda_\iota(b)}$. ∎

A correction at finitely many points of small weight does not change ν. We use this to discard the boundary terms in Corollary 15.15.

Lemma 15.13 (Finite corrections)**.** *Let K and L be constructible Weil complexes on an affine curve X_0, and let a be a real number. Suppose that there is a constant C with $\sum_{x \in X_0(k_n)} |\iota(t_{K,n} - t_{L,n})(x)|^2 \leq Cq^{na}$ for all n. If $\max(\nu(K), \nu(L)) > a-1$, then $\nu(K) = \nu(L)$.*

Proof. The inequality $|z+u|^2 \leq 2|z|^2 + 2|u|^2$, applied to $t_{L,n}$ and the difference $t_{K,n} - t_{L,n}$, gives $S_n(K) \leq 2S_n(L) + 2Cq^{na}$, so that $\nu(K) \leq \max(\nu(L), a-1)$, and by symmetry $\nu(L) \leq \max(\nu(K), a-1)$. If one of $\nu(K)$, $\nu(L)$ exceeds $a-1$, these two inequalities force equality. ∎

By counting points, Lemma lemma-bounds-squared-traces gives $\nu \leq w$ for a sheaf pure of weight w. A priori, however, the squared traces could be much smaller than this trivial bound, through cancellation among the eigenvalues at each point. The next theorem shows that on average they are not.

Theorem 15.14 (Weights and sums of squared traces)**.** *Let $X_0/\mathbf{F}_q$ be a smooth geometrically connected affine curve. For every nonzero pure lisse Weil sheaf $\mathcal{G}$ of weight w,*

$$\nu(\mathcal{G}) = w.$$

Proof. Lemma lemma-bounds-squared-traces gives $\nu(\mathcal{G}) \leq w$, so, by Lemma 15.11, we must show that the radius of $\varphi_{\mathcal{G}}$ is not larger than q^{-w-1}. For a real sheaf we find a pole of the logarithmic derivative of an L-function on that circle, and Proposition 15.9 reduces the general case to this one by twisting to weight zero and adding the dual.

The real case. Suppose first that $\mathcal{G}$ is real, and put $\mathcal{T} = \mathcal{G} \otimes \mathcal{G}$. The squared traces of $\mathcal{G}$ are then the traces of $\mathcal{T}$, and we express their generating series through the L-function of $\mathcal{T}$. Since $\iota t_{\mathcal{G},n}(x)$ is real, we have $\iota t_{\mathcal{T},n}(x) = |\iota t_{\mathcal{G},n}(x)|^2$ for $x \in X_0(k_n)$. The Euler product and the trace formula then give

$$\varphi_{\mathcal{G}}(t) = \frac{L'(t)}{L(t)}, \qquad L(t) = \frac{\iota \det(1 - Ft \mid H^1_c(X, \mathcal{T}))}{\iota \det(1 - Ft \mid H^2_c(X, \mathcal{T}))}.$$

For the first equality, group the local logarithms by degree. Then the coefficient of t^{n-1} in L'/L is the sum of the traces over $X_0(k_n)$, which is $S_n(\mathcal{G})$. As $\mathcal{T}$ is pure of weight $2w$, Lemma 14.20* shows that all roots of the denominator have absolute value $R = q^{-w-1}$.

We show that the function L has a pole on the circle $|t| = R$. This is not automatic, since a priori all the eigenvalues on H^2_c could cancel against eigenvalues on H^1_c, but positivity rules this out. Indeed, by Lemma 14.22*, each local Euler factor of $\mathcal{T}$ has nonnegative Taylor coefficients, so the coefficients of the full product dominate those of any one factor. Now choose a closed point of the curve. Since $\mathcal{G}$ has positive rank and Frobenius is invertible, the Euler factor of $\mathcal{T}$ at this point is the reciprocal of a nonconstant polynomial. It has infinitely many nonzero Taylor coefficients, because a polynomial cannot be inverse to a nonconstant polynomial. Hence the full Euler product is not a polynomial, and the rational function L has a pole, which by the bound on the denominator lies on $|t| = R$.

A pole of L is also a pole of its logarithmic derivative: if $L(t) = (t - t_0)^{-m} u(t)$ near t_0, with $m > 0$ and $u(t_0) \neq 0$, then $L'/L = -m/(t - t_0) + u'/u$. So the radius of convergence of $\varphi_{\mathcal{G}}$ is exactly R, and Lemma 15.11 gives $\nu(\mathcal{G}) = w$.

The general case. Let $\mathcal{G}$ be an arbitrary pure sheaf, and let $\mathcal{G}^0 = \mathcal{G} \otimes \mathcal{B}_b$ and $\mathcal{E}$ be as in Proposition 15.9. The sheaf $\mathcal{G}^0$ is pure of weight zero, and by Lemma lemma-bounds-squared-traces we have $\nu(\mathcal{G}^0) = \nu(\mathcal{G}) + \lambda_\iota(b) = \nu(\mathcal{G}) - w$, so it suffices to show that $\nu(\mathcal{G}^0) = 0$. The sheaf $\mathcal{E} = \mathcal{G}^0 \oplus (\mathcal{G}^0)^\vee$ is nonzero, real and pure of weight zero, so by the case already treated $\nu(\mathcal{E}) = 0$.

This lower bound passes to $\mathcal{G}^0$, because the trace of $\mathcal{E}$ is twice the real part of that of $\mathcal{G}^0$. In other words, for every n and $x \in X_0(k_n)$ we have $\iota t_{\mathcal{E},n}(x) = 2\,\mathrm{Re}(\iota t_{\mathcal{G}^0,n}(x))$. Hence $S_n(\mathcal{E}) \leq 4 S_n(\mathcal{G}^0)$, and so $\nu(\mathcal{G}^0) \geq 0$. Since the opposite inequality comes from Lemma lemma-bounds-squared-traces, we obtain $\nu(\mathcal{G}^0) = 0$, that is, $\nu(\mathcal{G}) = w$. ∎

Under the hypothesis of Proposition 16.17*, the Fourier transform of $(j_* M)[1]$ is again a middle extension. But Plancherel applies only to complexes on the whole line, boundary points included, so we need the theorem for j_* and $j_!$ on the larger curve. This costs nothing: the boundary points do not change ν, because their weights are small.

Corollary 15.15 (Sums of squared traces of $j_!$ and j_*)**.** *Let $j : U_0 \hookrightarrow X_0$ be a dense open of a smooth geometrically connected affine curve, and let $\mathcal{G}$ be a nonzero pure lisse Weil sheaf of weight w on U_0. Then*

$$\nu(j_! \mathcal{G}) = \nu(j_* \mathcal{G}) = w.$$

Proof. For $j_!$ the trace function is that of $\mathcal{G}$ extended by zero, and for j_* it differs from this at finitely many points of small weight, which Lemma 15.13 allows us to ignore.

The open curve U_0 is again smooth, geometrically connected and affine, and Theorem 15.14 on U_0 gives $\nu(\mathcal{G}) = w$. Extension by zero only adds zero traces, so $\nu(j_! \mathcal{G}) = w$. The trace functions of j_* and of the extension by zero differ only at the points of $X_0 - U_0$, by the trace function of the boundary stalks. These stalks have weights at most w by Proposition 15.5, so if i denotes the

inclusion of $X_0 - U_0$, then Lemma lemma-bounds-squared-traces, applied to the sheaf $i_*i^*j_*\mathcal{G}$, bounds the squared sum of the difference by Cq^{nw}. Since $\nu(j_!\mathcal{G}) = w > w - 1$, Lemma 15.13 shows that $\nu(j_*\mathcal{G})$ is still w. ■

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References marked * are to parts of the blueprint this document leaves out:

Corollary 10.13 Degreewise duality over fields, in II Duality and the trace formula (`corollary-degreewise-duality-over-fields`)

Theorem 12.24 Grothendieck–Lefschetz trace formula, in II Duality and the trace formula (`theorem-grothendieck-lefschetz-trace-formula`)

Definition 13.1 Weights, purity and mixedness, in III Weights and the Riemann hypothesis (`definition-weights`)

Example 13.4 The Kummer extension, in III Weights and the Riemann hypothesis (`example-kummer-extension`)

Proposition 13.7 Weights under the usual operations, in III Weights and the Riemann hypothesis (`proposition-weights-under-usual-operations`)

Lemma 14.20 Degree zero and top cohomology, in III Weights and the Riemann hypothesis (`lemma-degree-zero-top-cohomology`)

Definition 14.21 Real sheaves, in III Weights and the Riemann hypothesis (`definition-real-sheaves`)

Lemma 14.22 Positivity of Euler factors, in III Weights and the Riemann hypothesis (`lemma-positivity-euler-factors`)

Chapter 16 The Fourier transform, in III Weights and the Riemann hypothesis (`chap-The-Fourier-transform`)

Proposition 16.17 The Fourier transform of a middle extension, in III Weights and the Riemann hypothesis (`proposition-fourier-transform-middle-extension`)

Chapter 17 The Deligne–Laumon theorem for curves, in III Weights and the Riemann hypothesis (`chap-The-Deligne-Laumon-theorem-for-curves`)

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